

Thanh Nam Tran

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TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, C++

Frontend: React, Next.js, HTML, CSS, Tailwind CSS, Bootstrap

Backend and APIs: Node.js, Express.js, Django, Flask, FastAPI, Odoo ERP, REST APIs, WebSockets

Databases and infrastructure: PostgreSQL, Microsoft SQL Server, MongoDB, Supabase, AWS, Docker, Kubernetes, CI/CD, Git, Vercel

AI, ML, and data: PyTorch, scikit-learn, OpenCV, spaCy, LangChain, RAG, vector embeddings, model evaluation, pandas, NumPy, Power BI, Tableau

EXPERIENCE

Software Developer

02/2022–Present

Media Cybernetics, Inc.

Rockville, MD

- Build and maintain full-stack ERP applications in Odoo, including user interfaces, business logic, database workflows, integrations, and reusable modules.
- Support deployments and application reliability through CI/CD, Docker, Kubernetes, and performance monitoring. Migrated Odoo hosting from AWS to Odoo.sh in Q3 2026, reducing monthly AWS spend by 68%.
- Develop internal RAG and helpdesk chatbots and LLM-powered workflows for ticket summarization and resolution recommendations.

Engineering Intern

05/2021–12/2021

Media Cybernetics, Inc.

Rockville, MD

- Maintain and update legacy code to ensure compatibility between multiple legacy Capture Interface versions.
- Develop and improve Capture Interfaces features using C++, software SDK, and microscope cameras API.
- Built a company metrics dashboard using REST APIs, Microsoft SQL Server, Power BI, and batch workflows to automate reporting

PROJECTS

Jelly — Best First Time Hack, hackUMBC 2026

GitHub / Site / Devpost

Next.js, React, TypeScript, PostgreSQL, Drizzle ORM, OpenAI API

- Co-built Jelly in 24 hours with a two-person team, combining synthetic sportsbook data into a dashboard and personalized plans that help users track betting and pursue goals to limit, reduce, or stop gambling.
- Added an OpenAI-powered, CBT-inspired self-help chatbot informed by NICE gambling-harms guidance, with safeguards for crisis and betting-advice requests.

Share Your Song — Full-Stack Music Community Platform

GitHub / Site

Next.js, React, TypeScript, Tailwind CSS, MongoDB, NextAuth, AWS S3, Pusher, Resend

- A social music-sharing platform with secure user authentication, customizable profiles, media uploads, song and recommendation posts, comments, real-time chat, activity notifications, and transactional email workflows.

Adapting Vision GNN SAR Iceberg Imagery Classification

Report / Presentation

Vision Graph Neural Networks, Computer Vision, Model Fine-Tuning, Training, and Evaluation

- In a group research project, applied Vision Graph Neural Networks to classify ice, ocean, boundaries, and other regions in Antarctic SAR imagery using patch-level learning, self-supervised training, and graph-based feature analysis.

EDUCATION

University of Maryland, Baltimore County (UMBC)

01/2025–12/2026 (expected)

M.S. Computer Science – AI/ML Focus

Global Ambassador (Graduate Assistant), 01/2026–12/2026

Relevant Courses: Machine Learning, Principles of Artificial Intelligence, Neural Networks, Computer Vision

University of Maryland, Baltimore County (UMBC)

08/2019–12/2021

B.S. Computer Science – Data Science Focus